Szekvencia

$A = [1..6]$

S_1, S_2 programok A felett

$$S_1 = \left\{ \begin{array}{lll} 1 \to \langle 1,4,3 \rangle & 1 \to \langle 1,2,4 \rangle & 2 \to \langle 2,2,\dots \rangle \\ 2 \to \langle 2,1,4,6 \rangle & 3 \to \langle 3,5,1 \rangle & 4 \to \langle 4,5,3 \rangle \\ 5 \to \langle 5,1,\text{fail} \rangle & 6 \to \langle 6,3,1,5 \rangle & \end{array} \right\}$$

$$S_2 = \left\{ \begin{array}{lll} 1 \to \langle 1,3,2 \rangle & 1 \to \langle 1,2,4 \rangle & 2 \to \langle 2,6 \rangle \\ 3 \to \langle 3,4 \rangle & 4 \to \langle 4,\text{fail} \rangle & 4 \to \langle 4,5,1 \rangle \\ 5 \to \langle 5 \rangle & 6 \to \langle 6,4,3,2 \rangle & \end{array} \right\}$$

$$(S_1 ; S_2) =$$

$$= \left\{ \begin{array}{ll} 1 \to \langle 1,4,3,4 \rangle & 1 \to \langle 1,2,4,\text{fail} \rangle \quad 1 \to \langle 1,2,4,5,1 \rangle \\ 2 \to \langle 2,2,2,\ldots \rangle & 2 \to \langle 2,1,4,6,4,3,2 \rangle \quad 3 \to \langle 3,5,1,3,2 \rangle \\ 3 \to \langle 3,5,1,2,4 \rangle & 4 \to \langle 4,5,3,4 \rangle \quad 5 \to \langle 5,1,\text{fail} \rangle \\ 6 \to \langle 6,3,1,5 \rangle & \end{array} \right\}$$

$$\alpha \otimes \beta$$

Legyenek $\pi_1, \pi_2 \in A \to \mathbb{B}$, úgy, hogy

$$\pi_1 = \{(1, \text{igaz}), (2, \text{igaz}), (4, \text{igaz}), (5, \text{hamis}), (6, \text{hamis})\}$$

$$\pi_2 = \{(1, \text{igaz}), (2, \text{hamis}), (3, \text{igaz}), (4, \text{igaz}), (5, \text{hamis})\}$$

Határozd meg a $(\pi_1 : S_1, \pi_2 : S_2)$ vágatát!

1	2	3	4	5	6	A
1	2	3	4	5	6	π_1
I	I	—	I	H	H	
1	2	3	4	5	6	π_2
I	H	I	I	H	—	

$(\pi_1 : S_1, \pi_2 : S_2) =$

$1 \to \langle 1, 4, 3 \rangle$ $\quad 1 \to \langle 1, 2, 4 \rangle$ $\quad 1 \to \langle 1, 3, 2 \rangle$

$2 \to \langle 2, 2, \ldots \rangle$ $\quad 2 \to \langle 2, 1, 4, 6 \rangle$

$3 \to \langle 3, \text{fail} \rangle$ $\quad 3 \to \langle 3, 4 \rangle$ $\quad 4 \to \langle 4, 5, 3 \rangle$

$4 \to \langle 4, \text{fail} \rangle$ $\quad 4 \to \langle 4, 5, 1 \rangle$ $\quad 5 \to \langle 5, \text{fail} \rangle$

$6 \to \langle 6, \text{fail} \rangle$

Ciklus

$A = [1..5]$

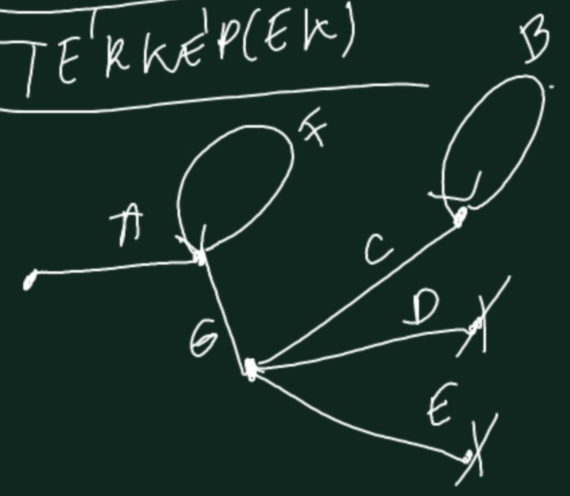
$$S_0 = \begin{cases} \overset{A}{1 \to \langle 1,2,4 \rangle} & \overset{B}{2 \to \langle 2 \rangle} & \overset{C}{3 \to \langle 3,4,2 \rangle} \\ \overset{D}{3 \to \langle 3,5 \rangle} & \overset{E}{3 \to \langle 3,3,\ldots \rangle} & \overset{F}{4 \to \langle \underline{4},5,3,\underline{4} \rangle} \\ \overset{G}{4 \to \langle 4,1,3 \rangle} & \overset{H}{5 \to \langle 5,5,5,\ldots \rangle} & \end{cases}$$

$\pi : A \to B$

$|\pi| = \{1,2,3,4\}$

$(\pi, S_0) = ?$

TÉRKÉP(EK)



$$(\Pi, S_0) = \Big\{$$

$a, b, c, d, e, f \in \mathbb{N}$

$1 \to \langle 1, 2, 4, (5, 3, 4)\infty \rangle$

$1 \to \langle 1, 2, 4, (5, 3, 4)a, 1, 3, 4, 2, 2, 2, \ldots \rangle$

$1 \to \langle 1, 2, 4, (5, 3, 4)b, 1, 3, 5 \rangle$

$1 \to \langle 1, 2, 4, (5, 3, 4)c, 1, 3, 3, 3, 3, \ldots \rangle$

$2 \to \langle 2, 2, 2, \ldots \rangle$

$3 \to \langle 3, 4, 2, 2, 2, \ldots \rangle$

$3 \to \langle 3, 5 \rangle$

$3 \to \langle 3, 3, 3, \ldots \rangle$

$4 \to \langle 4, (5, 3, 4)\infty \rangle$

$4 \to \langle 4, (5, 3, 4)d, 1, 3, 4, 2, 2, \ldots \rangle$

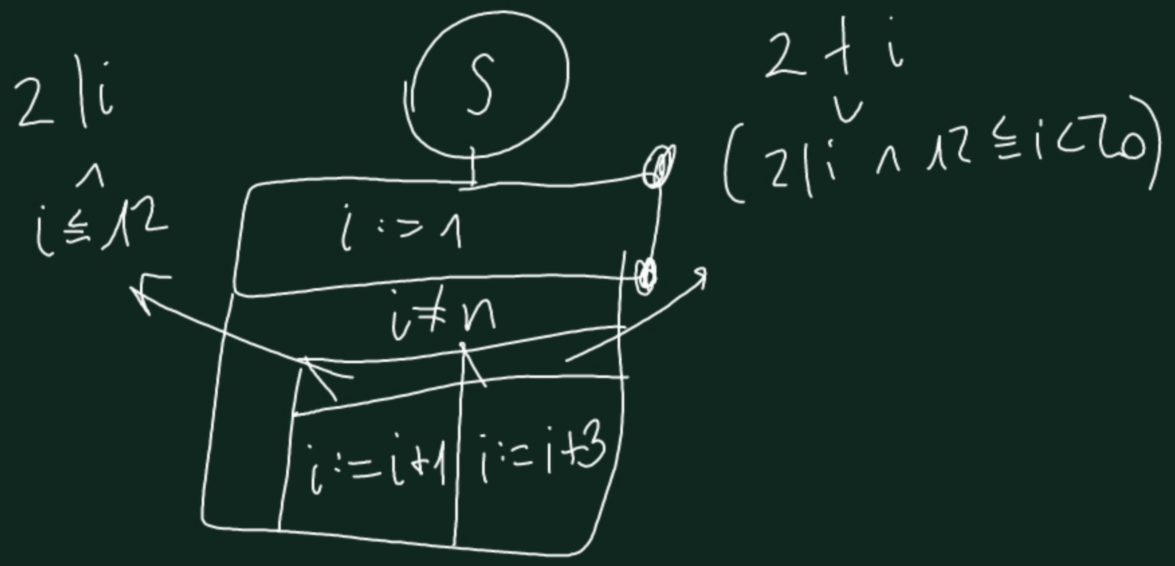
$4 \to \langle 4, (5, 3, 4)e, 1, 3, 5 \rangle$

$4 \to \langle 4, (5, 3, 4)f, 1, 3, 3, 3, \ldots \rangle$

$5 \to \langle 5 \rangle \Big\}$

$$A = (i : \mathbb{N},\ n : \mathbb{N})$$

$$S = \Big(i := 1;\ \mathrm{Do}\big(i \neq n,\ \mathrm{If}\big(2 \mid i \wedge i \leq 12 : i := i+1,\ 2 \nmid i \vee (2 \mid i \wedge 12 \leq i < 20) : i := i+3 \big)\big)\Big)$$



$(1,13) \longrightarrow$

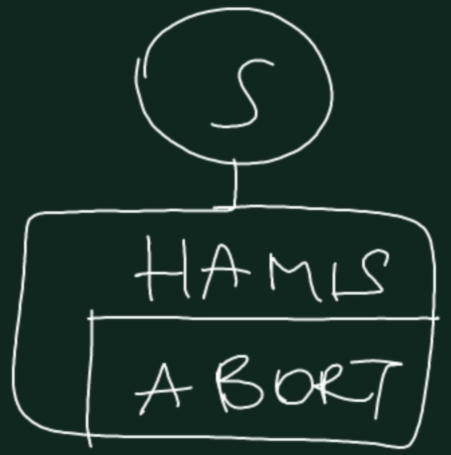
$\langle (1,13), (1,13), (4,13),$

$(5,13), (8,13), (9,13),$

$(12,13), \longrightarrow (13,13) \Rightarrow$

$\longrightarrow (15,13), (18,13), (21,13),$

$(24,13)$, fail



$$\forall a \in A : HAMIS(a) = hamis$$

$$S(HAMIS, ABORT)$$

$$\forall a \in A : S(a) = \{ \langle a \rangle \}$$

$$S = SKIP$$

ABORT